Karlsruher Institut für Technologie

Theoretical Investigation of Ethylene Oligomerization on Ni/Cr Single Site Catalysts on NU-1000

Thanh-Nam Huynh

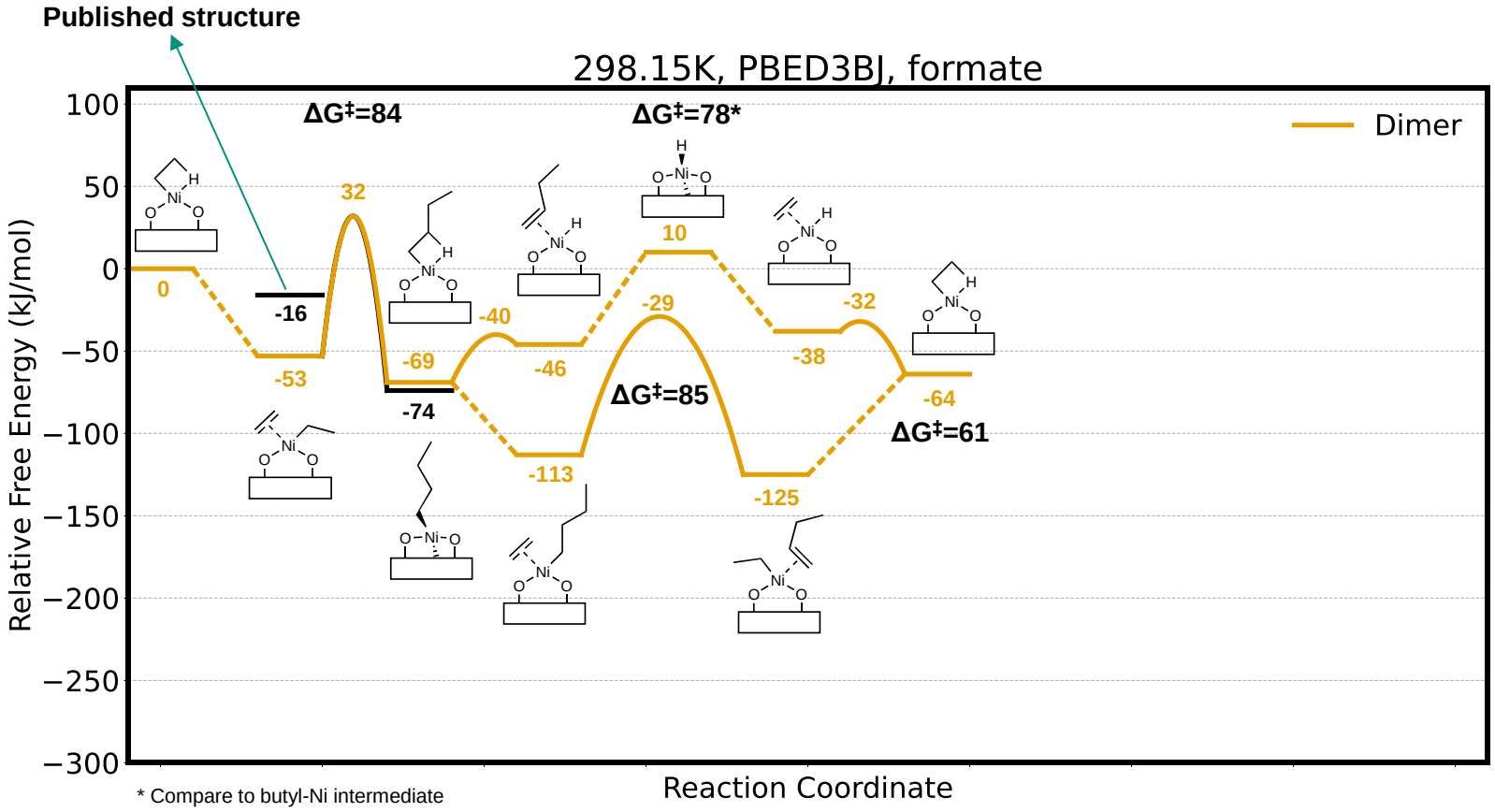
Bereich Theorie und Modellsysteme
Institut für Katalyseforschung und -technologie (IKFT)

P3 project meeting
29.04.2024

Outlines

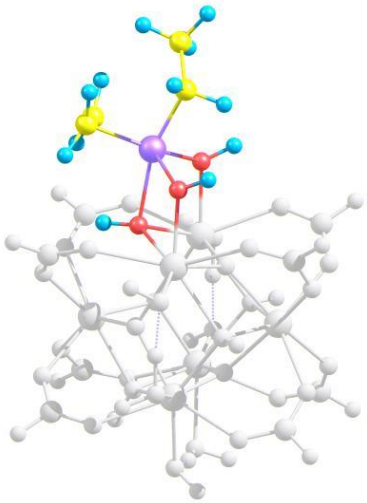
- Revised Mechanism for Ethylene Dimerization on Ni-NU-1000
- Ethylene Dimerization in the 3 Catalysts
- Ethylene Tri- and Tetramerization on Ni-NU-1000

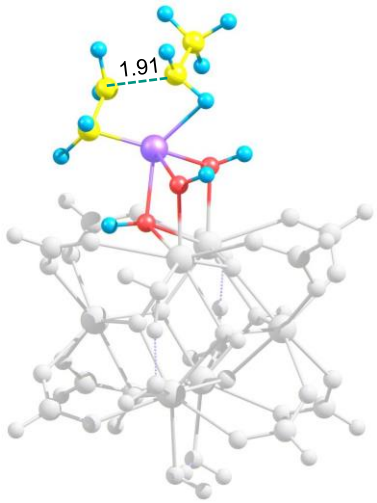
EtOligo in Ni-NU-1000

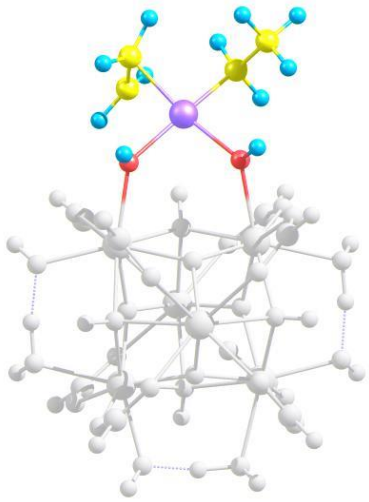


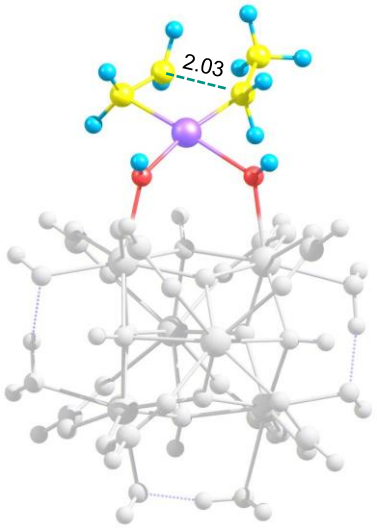
- C-C coupling/H-transfer TSs are the RDSs.
- Two but-1-ene desorption pathways: pressure-dependent?

EtOligo in Ni-NU-1000









Published structure

Less stable TS

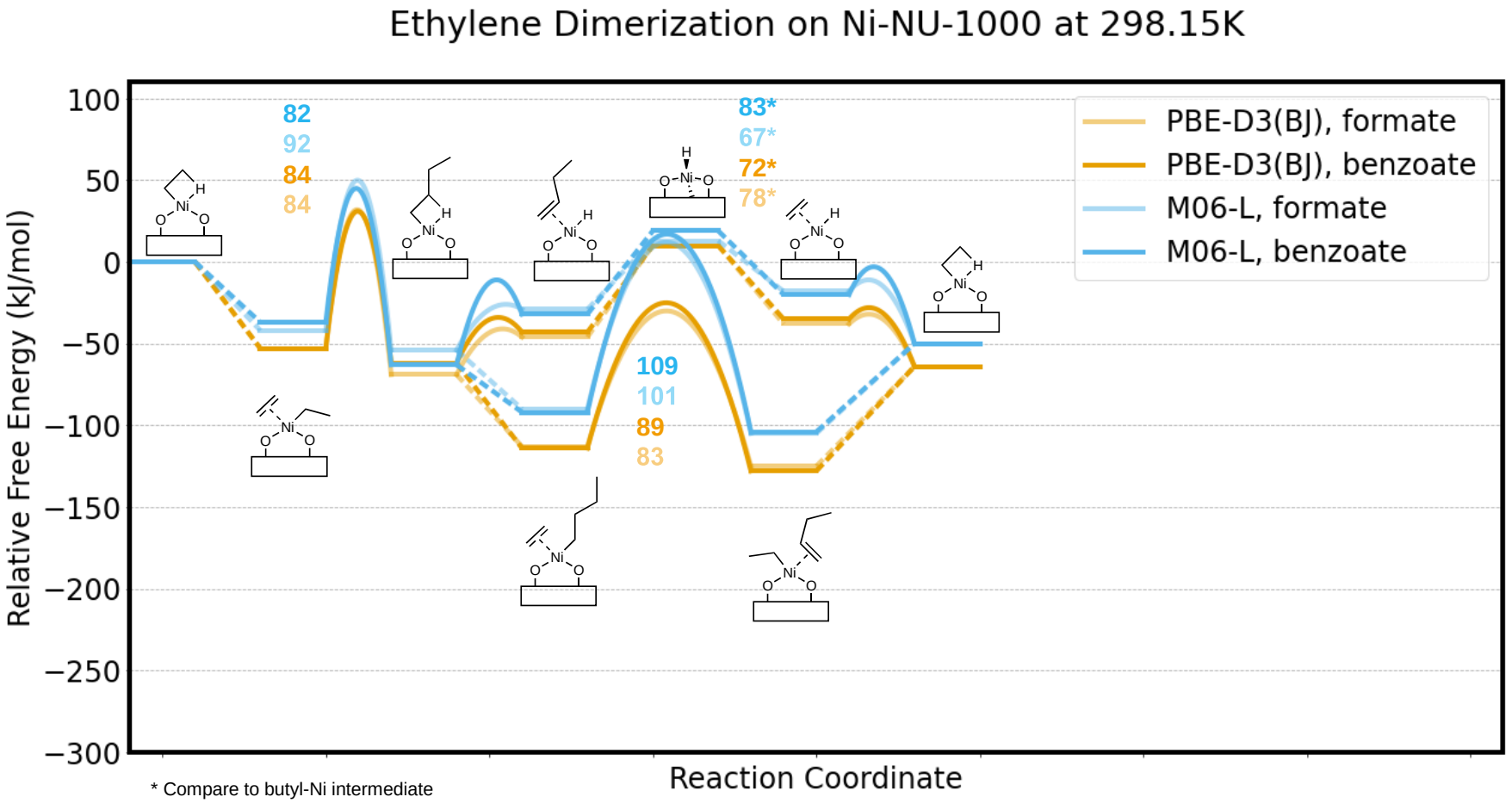
More stable structure (newly found)

Published TS (more stable)

$\Delta G^{\ddagger} = 98$

$\Delta G^{\ddagger} = 84$

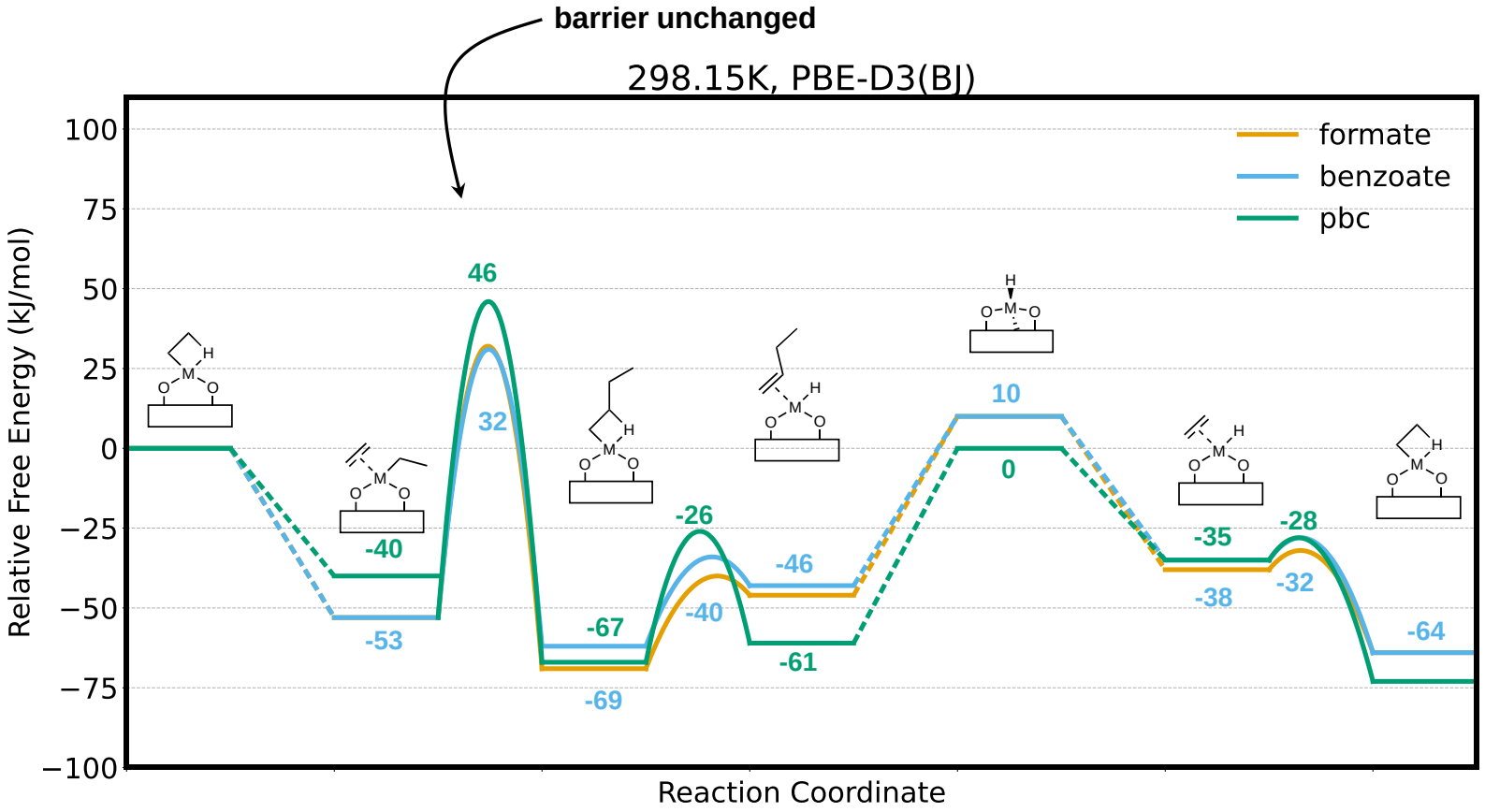
Ethylene Dimerization



* Compare to butyl-Ni intermediate

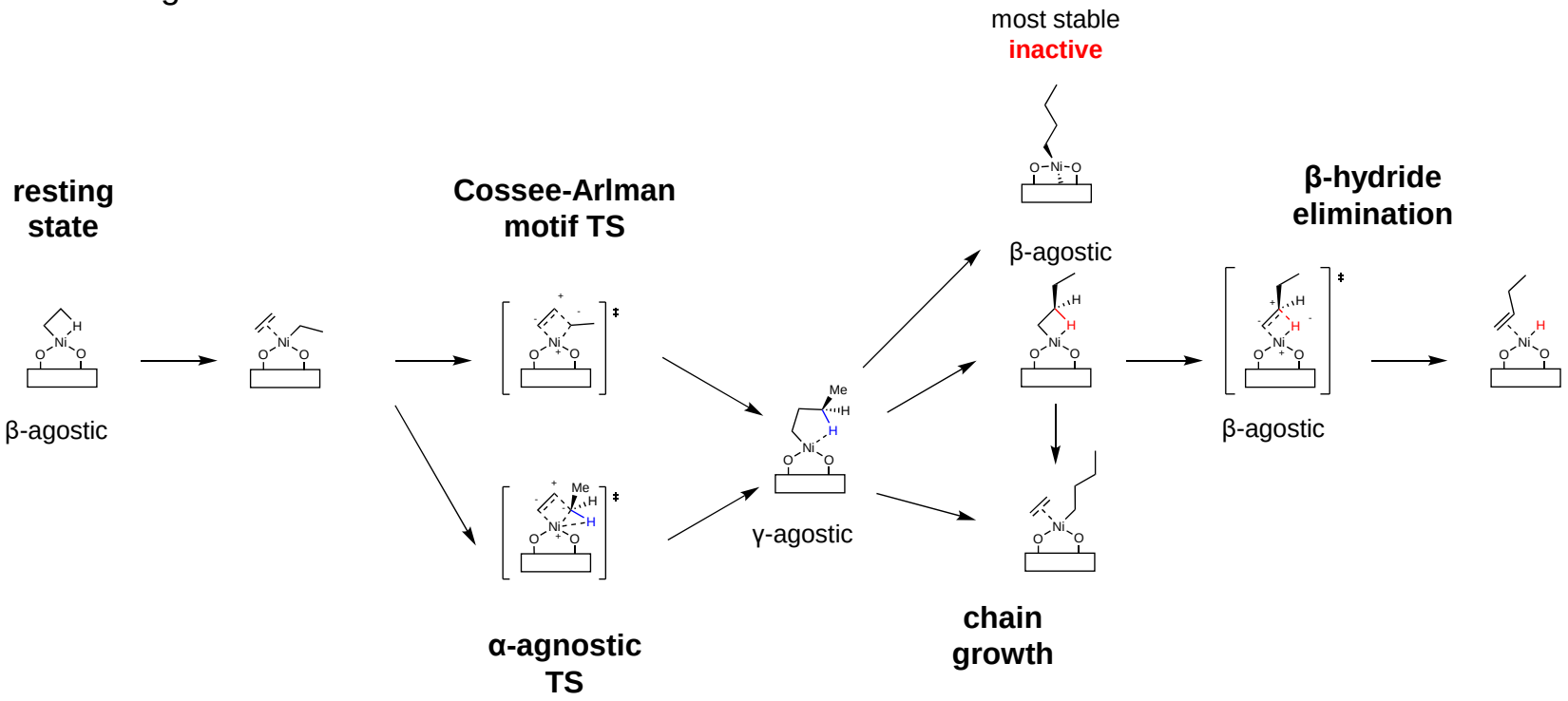
Ethylene Dimerization

- Steric effects



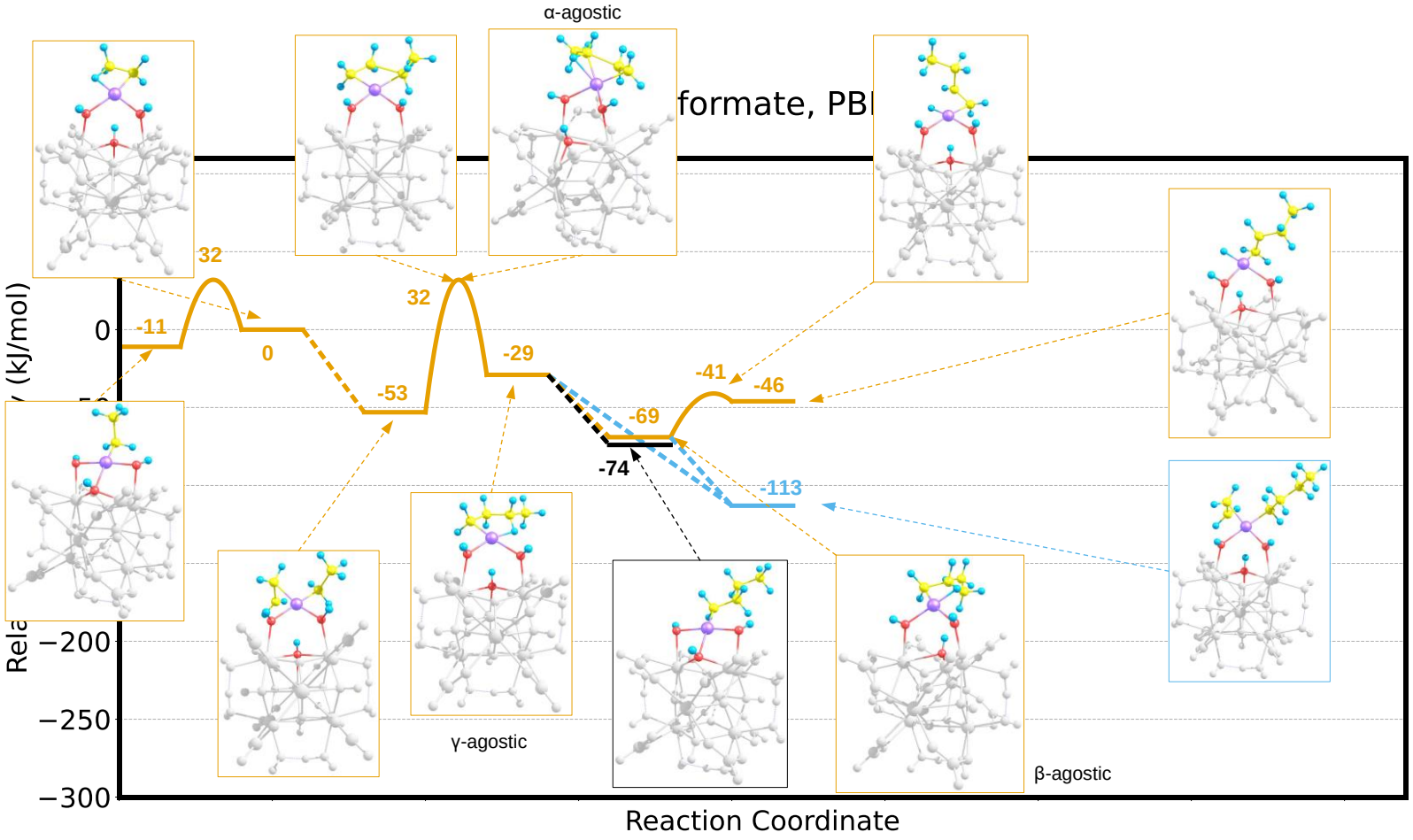
Dimerization – A detailed mechanism

- Role of agostic interactions

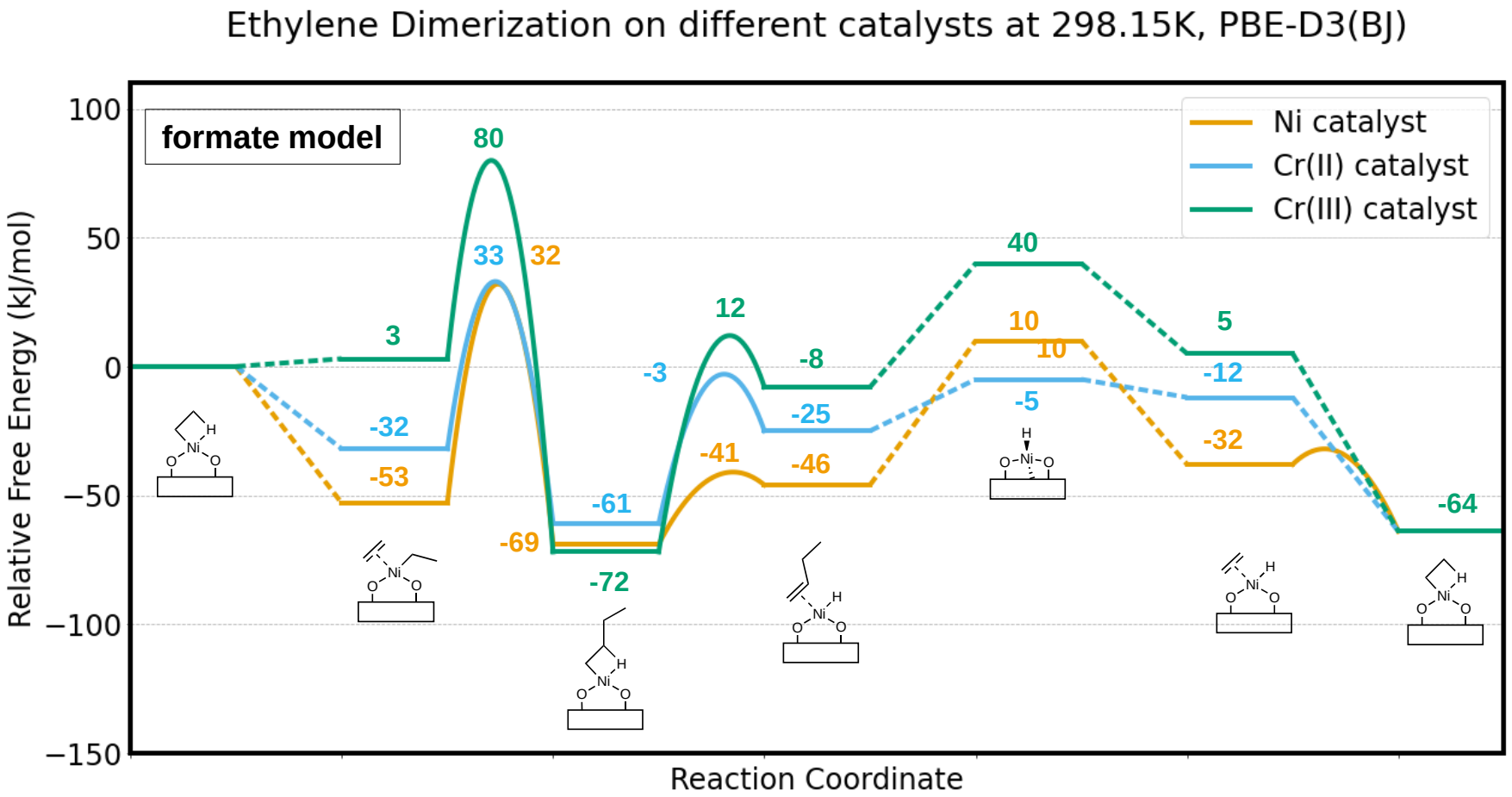


Dimerization – A detailed mechanism

- Role of agostic interactions



Ethylene Dimerization

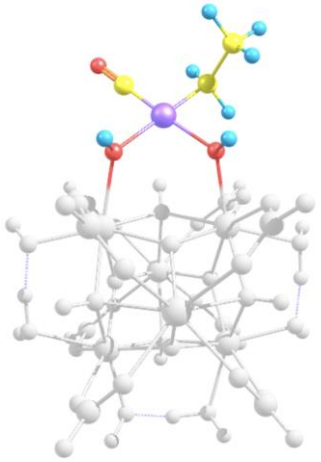


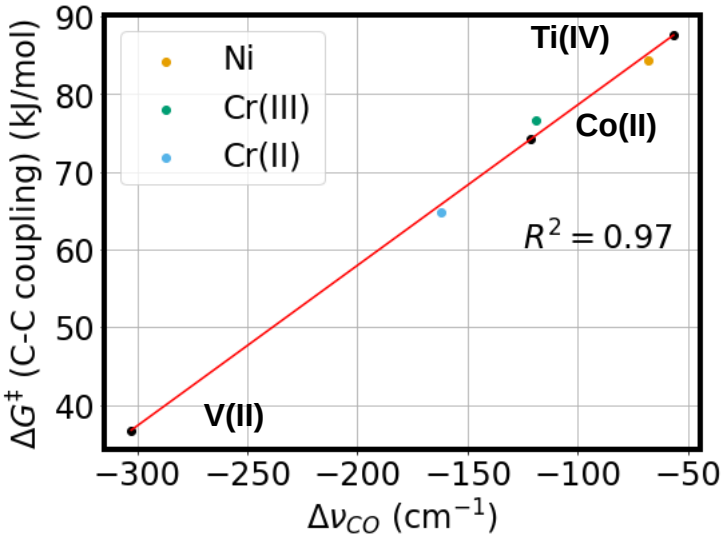
- Ethylene insertion is the RDS for Ni-NU-1000 and Cr(II)-NU-1000 (at PBE-D3(BJ) level)
- β-hydrogen elimination is the RDS for Cr(III)-NU-1000 -> favors polymerization.
- C-C coupling barriers decreases in the order of Ni < Cr(III) < Cr(II)

Ethylene Dimerization

- Correlation between red-shift in CO stretching freq. and ethylene insertion activation energies (?)

PBE-D3(BJ)	CO_g	CO-Ni-ethyl	CO-Cr(III)-ethyl	CO-Cr(II)-ethyl
ν_{CO}	2128.53	2060.62	2009.48	1966.6
C-C coupling barrier	-	84.38	76.68	64.78





PBE-D3(BJ)

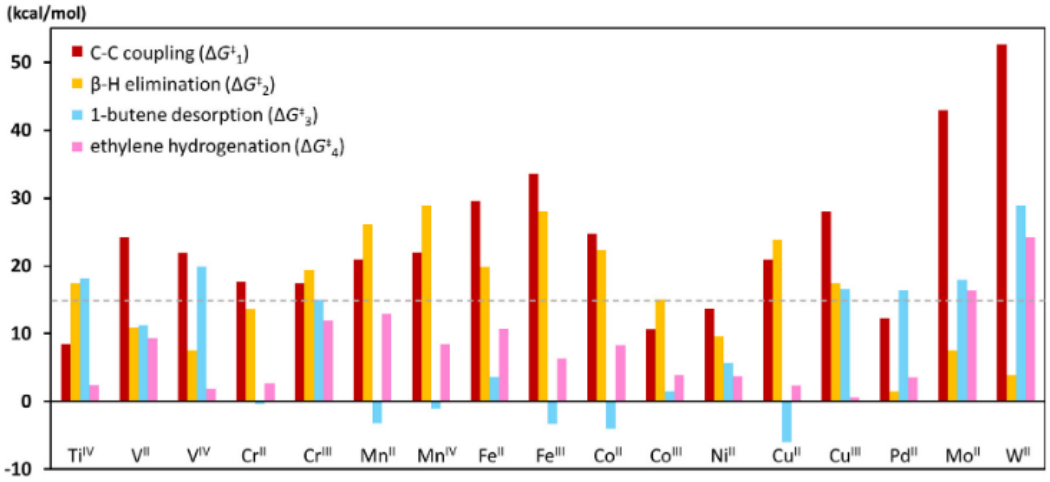
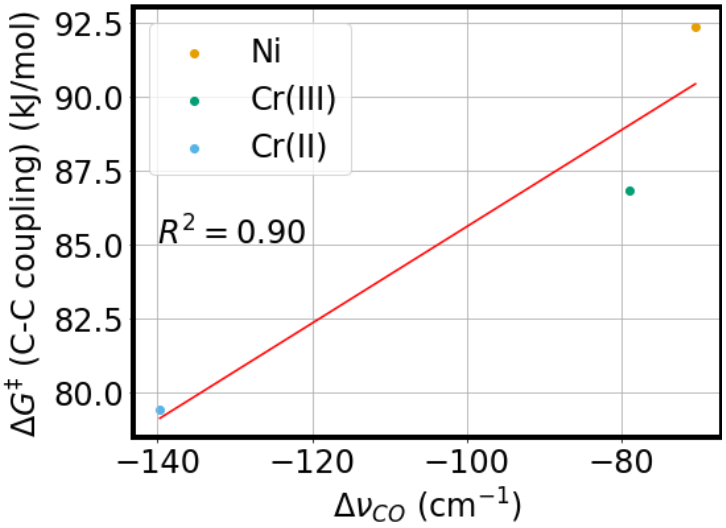


Fig. 5. The Gibbs free energies of activation for C—C coupling, β-H elimination, ethylene hydrogenation and 1-butene desorption.

Ethylene Dimerization

- Correlation between red-shift in CO stretching freq. and ethylene insertion activation energies (?)

M06-L	CO_g	CO-Ni-ethyl	CO-Cr(III)-ethyl	CO-Cr(II)-ethyl
ν_{CO}	2202.13	2131.68	2123.07	2062.46
C-C coupling barrier	-	92.36	86.81	79.39



M06-L

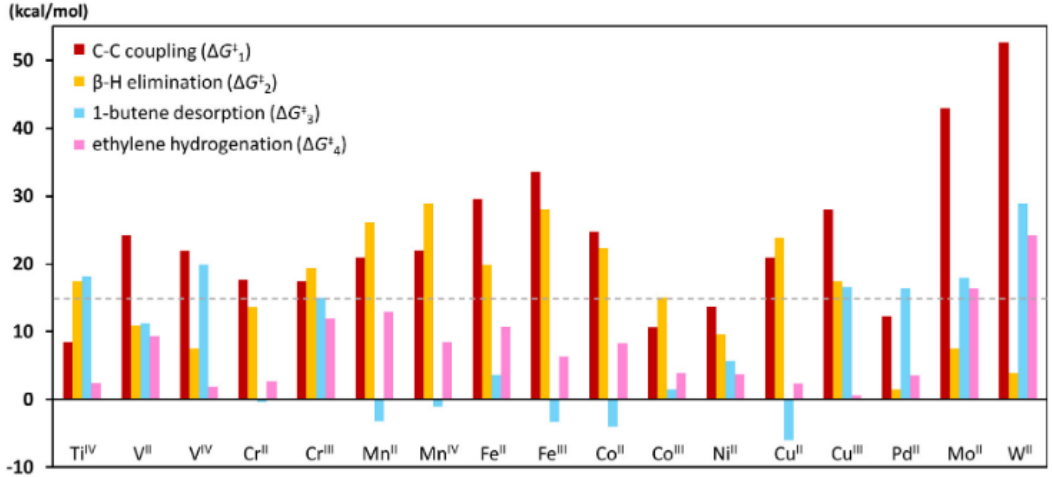
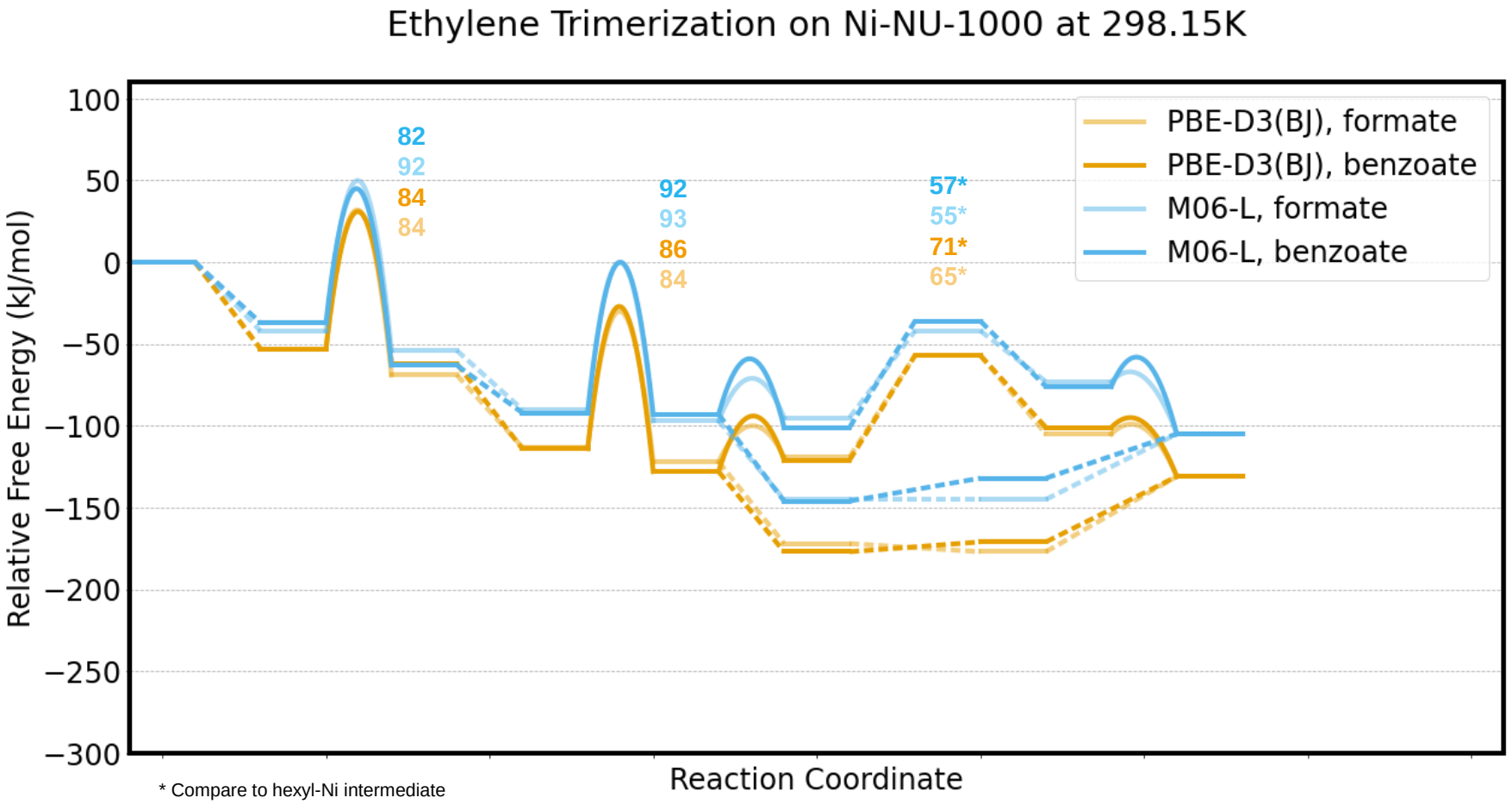


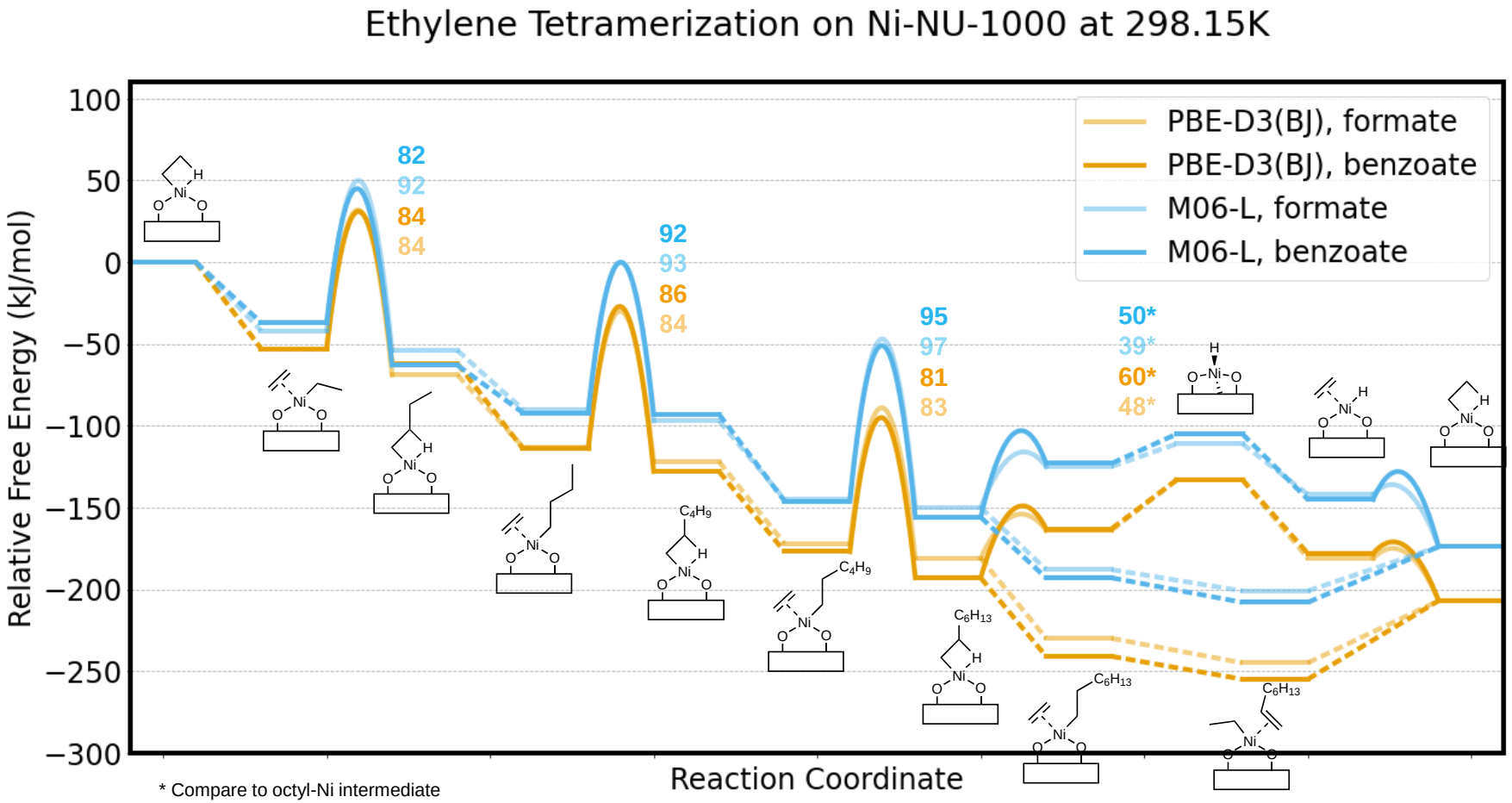
Fig. 5. The Gibbs free energies of activation for C—C coupling, β-H elimination, ethylene hydrogenation and 1-butene desorption.

Ethylene Trimerization



* Compare to hexyl-Ni intermediate

Ethylene Tetramerization



Ethylene Tetramerization

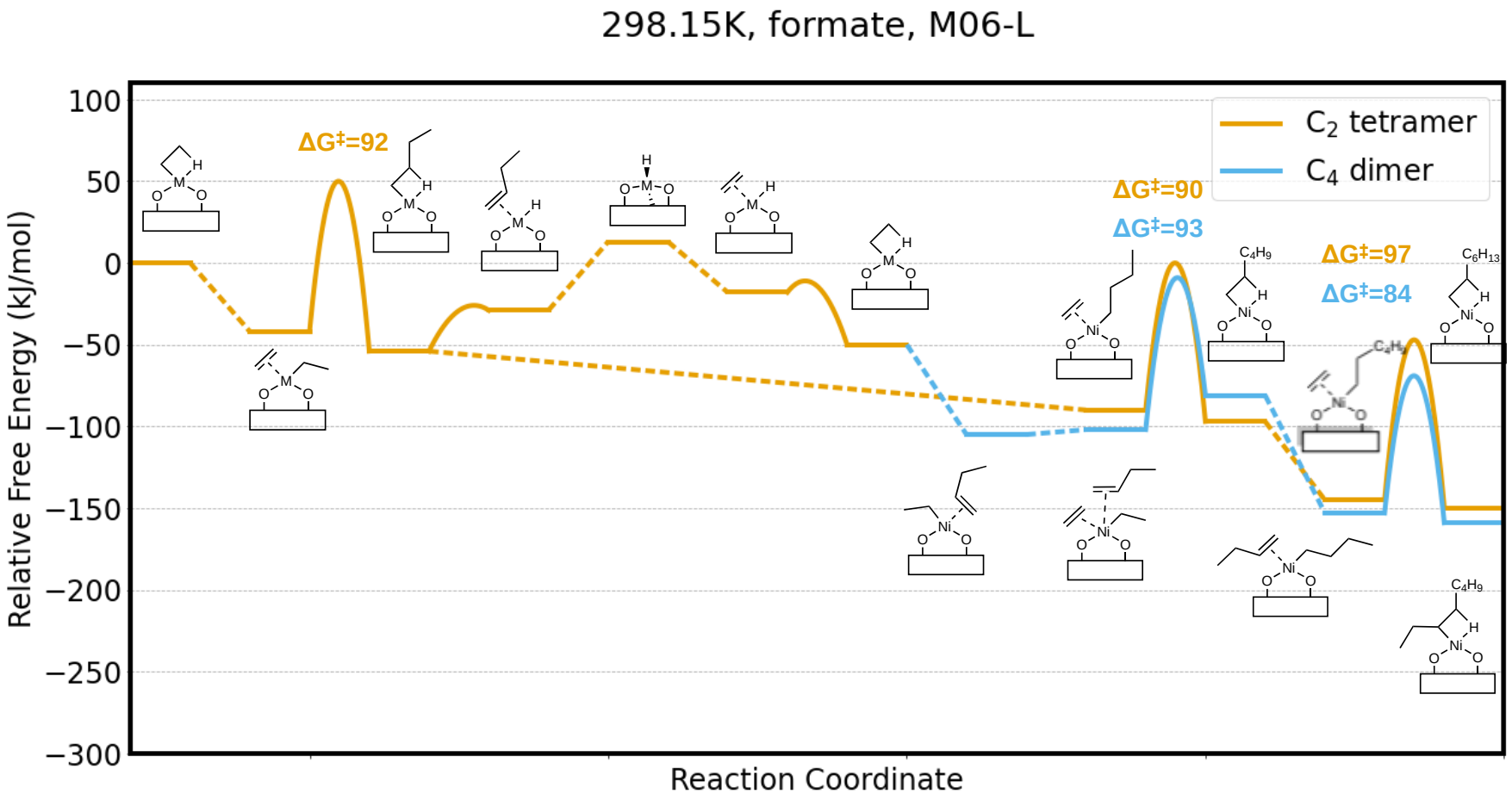
- Experimental data

Table S2 Product distribution of ethylene oligomerization catalyzed by activated Ni-AIM.

Ethylene Conversion (%)	C4 (%)	C6 (%)	C8 (%)
5	46	8	46
3	41	3	56
2	39	2	59

Ethylene Tetramerization

- Alternative pathway for octene formation (butene dimerization)



Conclusion and Outlook

- Dimerization:
  - Cr(II)-NU-1000 is predicted to have the lowest C-C coupling barrier (RDS)
  - Steric effects play little role in ethylene dimerization (at PBE-D3(BJ) level)
  - Ni-NU-1000: C-C coupling and butene desorption are RDSs (M06-L, benzoate)

- Tri- and tetramerization:
  - Steric effects influence the C-C coupling barrier at different lengths (M06-L)

- To meet experiments: a more favorable alternative pathway to C8 is required.